Curriculum Vitae

MD BILAYET HOSSAIN

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Mohishkundi-7052, Kushtia, Khulna, Bangladesh

EDUCATIONAL QUALIFICATIONS

Daffodil International University (DIU) **January 2022 -Present**
Bachelor of Science in Computer Science and Engineering *Dhaka-1216, Bangladesh*

- Current CGPA: 3.77 / 4.00 | Expected Graduation: January 2027

Peoples College , Khash Mathurapur **May 2018 – June 2020**
Higher Secondary Certificate (HSC) *Kushtia, Khulna, Bangladesh*

- GPA: 5.00 / 5.00 Field: Science

Pakuria Secondary School **January 2012 -February 2017**
Secondary School Certificate (SSC) *Kushtia, Khulna, Bangladesh*

- GPA: 5.00 / 5.00 Field: Science

INTERNATIONAL ACADEMIC EXPERIENCE

Academic Exchange Study Program (Japanese University Studies in Science & Technology (JUSST) Program).
The University of Electro-Communications, Tokyo, Japan
Duration: October 2024 – September 2025 (One Year)

- Successfully completed a one-year academic exchange program in Japan.
- Conducted research at the Yanai Laboratory under the supervision of Prof. Keiji Yanai.
- Participated in academic coursework, laboratory meetings, and collaborative research activities.
- Completed Japanese language training and intercultural academic programs.

RESEARCH PUBLICATIONS

I. Peer-Reviewed Publications

1. **Hossain, M.B.,** Yuan, H., Akhy, S.A., Yanai, K. (2026). "*BengaliDiff: Diffusion Model for Few-Shot Bengali Font Generation.*" In: Proceedings on Document Analysis and Recognition – ICDAR 2025 Workshops. ICDAR 2025, Wuhan, China. Lecture Notes in Computer Science, vol 16226. Springer, Cham.
   DOI: https://doi.org/10.1007/978-3-032-09371-4_7
2. Akhy, S.A., Shouno, H., Mojumdar, M.U., Hussain, T., **Hossain, M.B**., Siddiquee, S.M.T. and Chakraborty, N.R. (2025), MPICam: *A deep learning-based real-time medicinal plant identification system for rural healthcare in Bangladesh*. In: Proceedings of Distributed Computing, HPC, Cloud Computing, Quantum Technologies, Embedded Systems, Modeling, and Simulation (CSCE 2025), Las Vegas, USA, 21–24 July.
   DOI : https://doi.org/10.1007/978-3-032-22193-3_41

I I. Manuscripts Under Review / Submitted

1. Bhowmik, P., Rafi, M.A.H., Talukder, M.A., Shrabon, M.S.I., **Hossain, M.B.**, and Siam, M.M.A. (2026). "*Multi-Paradigm XAI with Lightweight Explainable Convolutional Network and Feature Representation for Waste Classification*." Submitted to the *Turkish Journal of Electrical Engineering & Computer Sciences*.
   Status: Submitted; Manuscript ID: TJEECS-2026-00985.

2. Anis, S.M., Mubin, M.M.R., Khandoker, M.A., **Hossain, M.B**., Bhuiyan, M.N., Nur, F.N., Rashel, M.R., and Noori, S.R.H. (2026). "*Difficulty-Aware Cost-Constrained Routing of Vision-Language Models for Efficient Visual Question Answering*." Submitted to the 29th International Conference on Computer and Information Technology (ICCIT 2026), Cox's Bazar, Bangladesh. Status: Submitted; Paper ID: 1995.

RESEARCH EXPERIENCE

Research Assistant (RA) **April 2026- Present**
Health Informatics Research Lab (HIRL)**,**
Daffodil International University, (DIU) Bangladesh
Research Activities:

- Conducting research in computer vision and health informatics, focusing on CNN- and Vision Mamba-based medical image analysis, model evaluation, explainable AI, experimental analysis, and research manuscript preparation.

Individual Research Project - Yanai Laboratory
The University of Electro-Communications, Tokyo, Japan
Title: BengaliDiff: Diffusion Model for Few-Shot Bengali Font Generation
Supervisor*: Prof. Keiji Yanai*
Duration: October 2024 – September 2025 (One Year)

- Conducted research on few-shot Bengali font generation by developing a diffusion-based framework with content style representation learning under the JUSST academic exchange program.
- Participated in Laboratory research meetings, technical discussions, and collaborative research activities.
- Research resulted in a peer-reviewed publication in the ICDAR 2025 Workshops (Springer LNCS) and was presented at ICDAR 2025, Wuhan, China.

SOFTWARE AND TECHNICAL KNOWLEDGE

- **Programming:** Python, C, C++, JavaScript , Dart, Bash
- **Web & Software Development:** HTML, CSS, React.js , TypeScript
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, MATLAB
- **Tools & Platforms:** Git, TensorBoard, VS Code, CodeBlocks, Android Studio, Kaggle, Google Colab.
- **Research & Writing:** LaTeX (Overleaf), Zotero, Mendeley
- **Operating Systems:** Linux, Windows

UNDERGRADUATE THESIS (B.Sc. ENGINEERING RESEARCH) (February 2026 – Present)

Title: *SpectraMoE-MambaNet: A Lightweight Spectral-Spatial Mamba Network with Adaptive Expert Routing for Medical Image Segmentation*

Supervisor: Dr. Md. Fokhray Hossain
Dean, Faculty of Science & Information Technology (FSIT) & Professor, Department of Computer Science & Engineering (CSE), Daffodil International University (DIU), Bangladesh
Research Focus: Developing SpectraMoE-MambaNet, a lightweight spectral-spatial Vision Mamba framework with adaptive expert routing for medical image segmentation, integrating efficient feature fusion and cross-gated attention, with a planned diffusion prior to improve lesion boundary refinement, segmentation robustness, and reliability while preserving computational efficiency.

CERTIFICATIONS & ACHIEVEMENTS

A. International & Academic Programs

• International Exchange Study program "The Japanese University Studies in Science & Technology (JUSST)", -

The University of Electro-Communications, Tokyo, Japan| 2024 – 2025

- Certificate of Commendation for Research Publications, Yanai Laboratory, UEC, Japan | 2025
- Certificate of Japanese Language Course "Elementary Japanese 1A" UEC, Japan | 2025
- Certificate of Global Healthcare Leadership Program, Daegu Health College, South Korea | 2022

B. MOOC, Workshop, and Competitions

- 1st Runner-up of "ModalX-AI Challenge – "AI-Powered Multi-Modal Presentation Assessment System", Daffodil International University (DIU) | January 2026
- 1st Runner-up of "Research Poster Presentation" at 5th National Data Science Summit - DIU 2026
- Certificate of Appreciation – Mentor, "AI Innovation Hackathon 2026 " | July 2026
- Certificate of Participation, "AI Dev Fest 2026 - Datathon", Daffodil International University | October 2026
- Certificate of Participation, "DIU-DoR Poster Presentation 2026", Daffodil International University | May 2026
- Completed "AI Engineering Masterclass" Course offered by Department of CSE, DIU (April – August 2026)
- Completed Bootcamp on "Computer Vision & Deep Learning for Medical Data Analysis", Daffodil International University (DIU), (HIRL Lab) | December 2025

SELECTED RESEARCH & UNDERGRADUATE PROJECTS

1. BengaliDiff: Diffusion Model for Few Shot Bengali Font Generation (October, 2024 – Present)

Technologies: Python, PyTorch, Diffusion Models, Computer Vision

- Developed a diffusion-based few-shot Bengali font generation framework with content style representation learning and adversarial supervision.
- Published in ICDAR 2025 Workshops (Springer LNCS). Paper Link (GitHub: BengaliDiff)

2. XAI-Driven Clinical Decision Support System for Interpretable Anemia Classification (April 2026- Present)

Technologies: Machine Learning, Random Forest, SHAP, Explainable AI, React, TypeScript

- Developed an explainable AI-based clinical decision support system for anemia classification using hematological parameters, achieving **98.98% accuracy** with Random Forest and integrating SHAP-based interpretation. (GitHub: Anemia Classification) & Live Demo

***** Achieved **1st Runner-Up in the Research Poster Presentation at the 5th National Data Science Summit, DIU 2026**.

3. ModalX-AI: Offline Video-Based Sentiment and Presentation Analysis System (March 2026 – Present)

Technologies: Python, Artificial Intelligence, Computer Vision, Video Analysis

- Developed an offline AI system for analyzing recorded presentation videos and generating sentiment and presentation-performance assessments while preserving user privacy through local processing.

***** Achieved **1st Runner-up** in the "**ModalX-AI Challenge – AI-Powered Multi-Modal Presentation Assessment System**", Daffodil International University, January 2026. (GitHub: ModalX-AI)

4. Dhaka Tesla Pool – Ride Sharing System (September 2026 – Present)

Technologies: React, TypeScript, Vite, Node.js, Express.js, PostgreSQL, Docker, Vitest

- Developed a Dockerized full-stack ride-sharing platform using React, Node.js, PostgreSQL, secure session-based authentication, and role-based access control. (GitHub: Dhaka Tesla Pool)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Executive Member, DIU Computer Programming Club (CPC) *(February, 2023–2024)*

• ***Responsibilities:*** Supported programming-related activities, technical events, peer learning, and knowledge-sharing initiatives among students.

Executive Member, Daffodil AI Club *(December, 2025– September 2026)*

• ***Responsibilities:*** Contributed to AI-focused workshops, research discussions, technical events, team coordination, and collaborative learning activities.

REFERENCES

Professor Dr. Md. Fokhray Hossain

Designation: Dean & Professor
Department: Computer Science & Engineering
Faculty: Faculty of Science and Information Technology
Daffodil International University (DIU), Bangladesh
Relationship: Undergraduate Thesis Supervisor
Email: drfokhray@daffodilvarsity.edu.bd
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Dr. Md. Zahid Hasan

Designation: Associate Professor
Department: Computer Science & Engineering
Faculty: Faculty of Science and Information Technology
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Email: zahid.cse@diu.edu.bd
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